MULTI-LOCATION INVENTORY ALLOCATION & PURCHASING INTELLIGENCE SYSTEM

Project Type: Inventory Management, Data Analysis & Business Reporting
Tools Used: Microsoft Excel, XLOOKUP, SUMIFS, IF, MIN, MAX, Data Validation, Charts and Dashboard Reporting

1. Project Overview

This project involved developing an Excel-based inventory allocation and purchasing decision-support system to help a business manage stock across multiple locations.

The objective was to connect customer orders, inventory records, stock reservations, incoming shipments, and supplier information to improve inventory allocation decisions, identify shortages, and support purchasing recommendations.

2. Data Preparation and Organization

I organized the data into separate Excel worksheets, including SalesOrders, Inventory, Reservations, Allocations, PurchaseOrders, OnTheWater, Locations, and Suppliers.

Each worksheet represented a different part of the business operation.

I used common identifiers such as product IDs, order IDs, inventory IDs, and supplier IDs to connect information across the worksheets.

Purpose: To create an organized data structure that allows information from different business activities to work together.

3. Inventory Availability Analysis

I calculated the quantity of stock available for new customer orders using the Available-to-Promise (ATP) method.

The calculation considered:

- Stock currently on hand.
- Quantities already committed.
- Minimum stock levels that must be protected.
- Active stock reservations.

I also introduced an inventory review check to flag stock records requiring verification before they could be used for new allocation recommendations.

Purpose: To prevent the business from promising inventory that may already be committed or unavailable.

4. Customer Order Analysis

I analyzed confirmed customer orders to determine which products were requested, the quantities ordered, and the required delivery dates.

I connected the SalesOrders worksheet to reservation records to identify quantities already set aside for customers.

I then calculated the additional quantity each order still required.

Purpose: To establish the exact outstanding demand before recommending additional inventory.

5. Multi-Location Inventory Allocation

I developed allocation formulas to recommend how available inventory could be distributed across multiple locations to fulfill customer orders.

The model considered available stock, outstanding customer demand, and quantities already proposed from each inventory source.

I used Excel formulas to prevent the same stock from being proposed to multiple orders and to ensure proposals did not exceed customer requirements.

Purpose: To distribute available inventory efficiently while reducing the risk of over-allocation.

6. Delivery Deadline and Product Validation

I connected inventory sources to their locations and retrieved estimated delivery times.

I calculated expected arrival dates and compared them with customers' required delivery dates.

I also introduced product-matching checks to confirm that each proposed inventory source contained the correct product.

The model rejects proposed quantities when the product does not match or the estimated delivery date is too late.

Purpose: To ensure inventory recommendations consider both product accuracy and delivery feasibility.

7. Inventory Shortage Analysis

After generating allocation proposals, I calculated the quantities that could not be covered by available inventory.

I summarized shortages by product to identify which items required additional attention.

Key findings:

Product	Proposed Supply	Unfulfilled Quantity
SKU-A	60	30
SKU-B	30	0
SKU-C	18	2
Total	**108**	**32**

SKU-A accounted for 30 of the 32 unfulfilled units, making it the highest-priority product for replenishment review.

8. Incoming Inventory and Purchase Order Analysis

I analyzed existing purchase orders and shipments in transit to determine whether incoming inventory could cover customer shortages.

I separated stock that had been ordered but not yet shipped from stock already in transit.

I compared expected shipment arrival dates with customer deadlines to identify which incoming quantities could potentially support fulfillment.

Purpose: To avoid unnecessary purchasing and prevent the same incoming stock from being counted twice.

9. Supplier Lead-Time and Purchasing Analysis

I compared suppliers using their production lead times, shipping durations, customs processing times, and domestic delivery periods.

I calculated total replenishment lead times and estimated arrival dates for different supplier options.

The analysis identified:

Supplier	Total Lead Time
SUP-US	8 days
SUP-EU	15 days
SUP-AS	37 days

SUP-US had the shortest modeled replenishment time.

The purchasing model compared these lead times with customer deadlines to identify potentially feasible options and flag cases requiring expedited supply or further review.

Purpose: To support purchasing decisions based on delivery requirements, rather than supplier selection by speed alone.

10. Dashboard Development

I developed an Excel dashboard to communicate inventory performance and purchasing priorities.

The dashboard included key performance indicators, a product-level shortage summary, a comparison chart, and business insights.

Dashboard results:

Key Performance Indicator	Result
Total Confirmed Demand	150 units
New Allocation Proposals	108 units
Unfulfilled Demand	32 units
Provisional Fulfillment Coverage	78.67%

The dashboard was connected to the underlying Excel calculations, allowing the figures to update when order quantities or allocation proposals changed.

Purpose: To help management understand inventory availability, fulfillment gaps, and purchasing priorities at a glance.

11. Testing and Quality Checks

I tested the model by changing order quantities, delivery deadlines, and product identifiers to observe how the calculations responded.

The tests demonstrated that the system could:

- Update inventory and allocation calculations when quantities changed.
- Prevent repeated use of the same stock within the allocation proposal model.
- Reject proposed allocations that failed product-matching checks.
- Reject stock that could not meet the modeled delivery deadline.
- Recalculate shortages and purchasing recommendations.

I also introduced review flags and documented an existing discrepancy between historical allocation and reservation records.

Purpose: To improve calculation reliability and identify records requiring further verification.

12. Key Results and Business Insights

The analysis produced the following findings:

- Evaluated 150 units of confirmed customer demand.
- Generated 108 units of additional inventory allocation proposals.
- Identified 32 units of outstanding demand.
- Achieved a provisional coverage calculation of 78.67%, including existing reservations and new allocation proposals.
- Identified SKU-A as the primary product requiring replenishment attention.
- Compared supplier lead times to identify possible purchasing alternatives.
- Developed a dynamic dashboard for inventory monitoring and reporting.

13. Conclusion

This project demonstrates how Microsoft Excel can be used to connect business data, automate inventory analysis, evaluate fulfillment options, and support purchasing decisions.

By combining inventory availability calculations, customer order analysis, reservation tracking, allocation controls, delivery validation, and supplier comparisons, I developed a structured decision-support model that turns operational records into useful business insights.

The project strengthened my skills in Excel formulas, data management, operational analysis, problem-solving, and business reporting.

Project Scope: Independent portfolio case study using constructed business data. Allocation figures are proposed quantities rather than completed transactions. Delivery estimates use simplified assumptions, and documented historical allocation discrepancies require review before operational use.